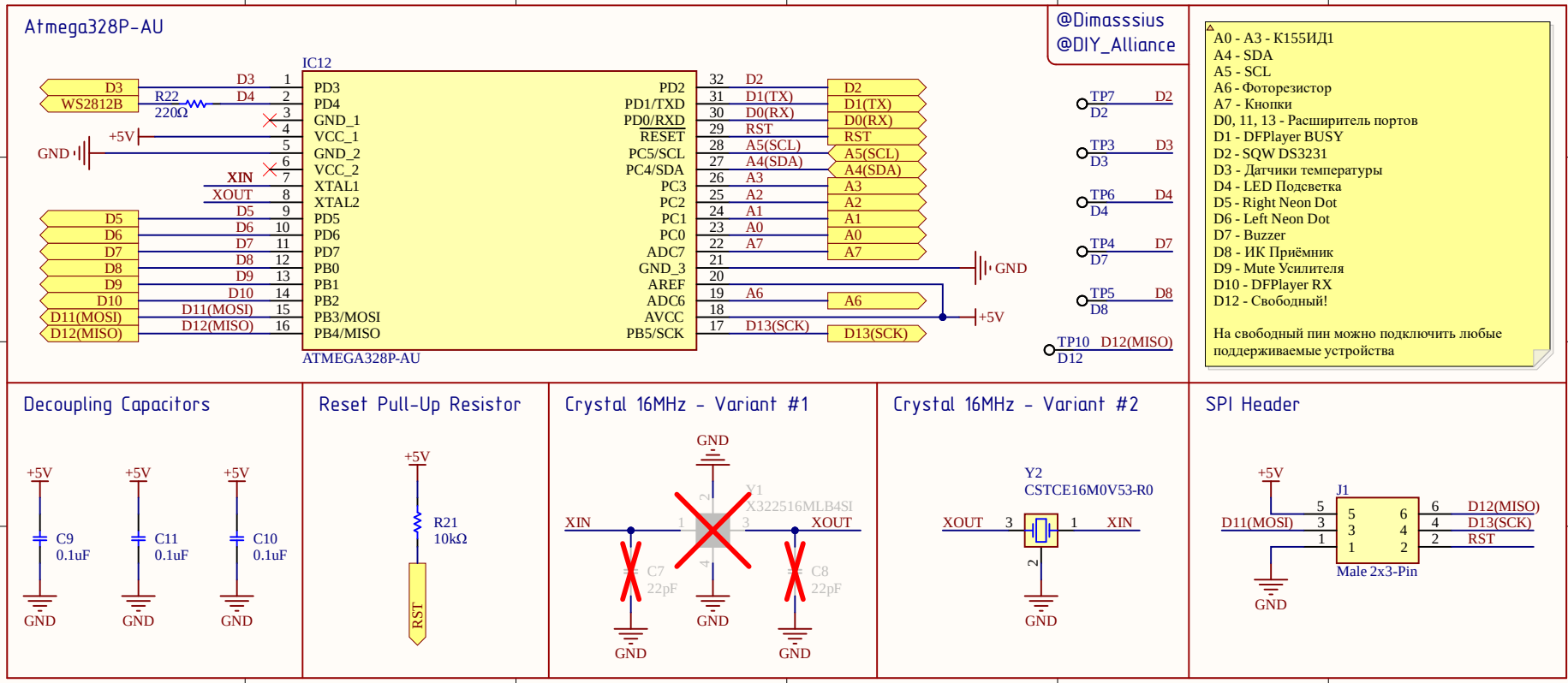
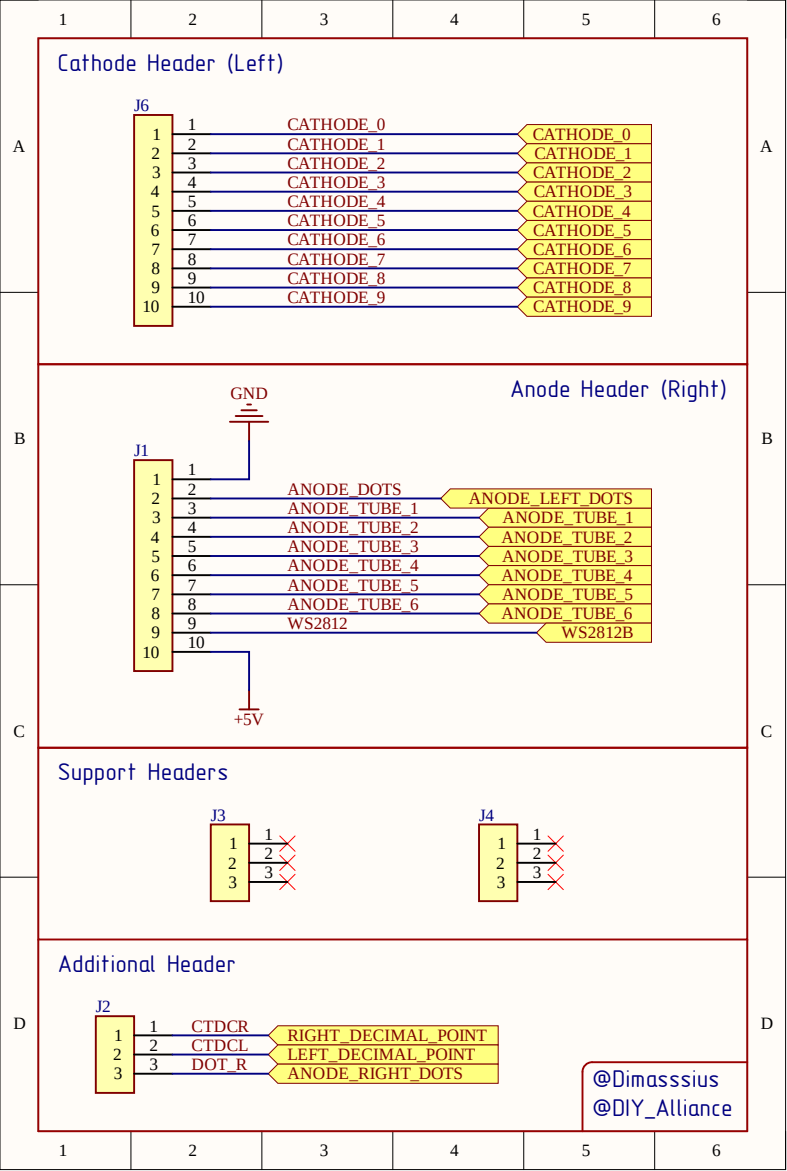
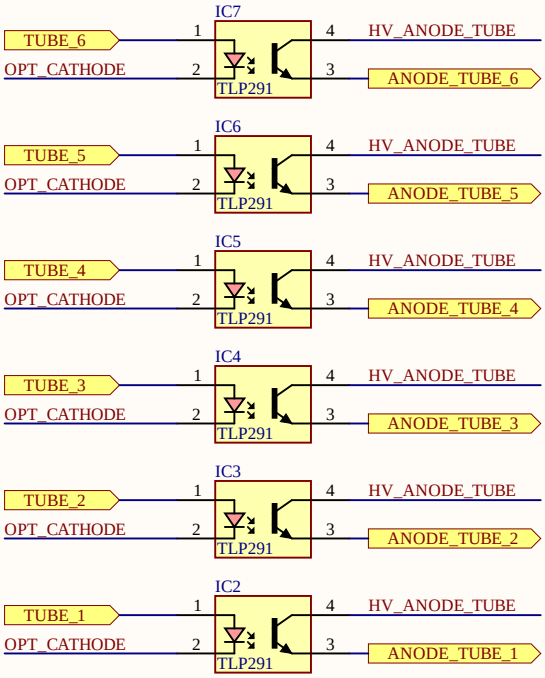
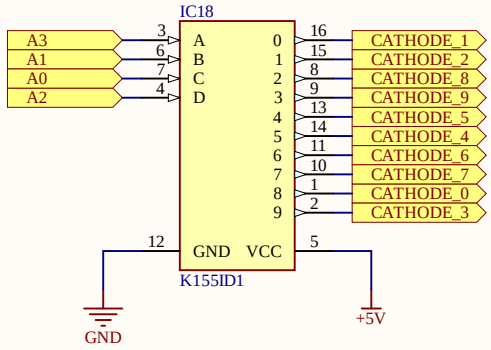

Atmega328P-AU
IC12
ATMEGA328P-AU
@Dimasssius
@DIY_Alliance
A0 - A3 - К155ИД1
A4 - SDA
A5 - SCL
A6 - Фоторезистор
A7 - Кнопки
D0, 11, 13 - Расширитель портов
D1 - DFPlayer BUSY
D2 - SQW DS3231
D3 - Датчики температуры
D4 - LED Подсветка
D5 - Right Neon Dot
D6 - Left Neon Dot
D7 - Buzzer
D8 - ИК Приёмник
D9 - Mute Усилителя
D10 - DFPlayer RX
D12 - Свободный!
На свободный пин можно подключить любые поддерживаемые устройства
Decoupling Capacitors
C9 0.1uF
C11 0.1uF
C10 0.1uF
Reset Pull-Up Resistor
R21 10kΩ
RST
Crystal 16MHz - Variant #1
Y1 X322516MLB4SI
C7 22pF
C8 22pF
Crystal 16MHz - Variant #2
Y2 CSTCE16M0V53-R0
SPI Header
J1
Male 2x3-Pin


Cathode Header (Left)
J6
1 CATHODE_0
2 CATHODE_1
3 CATHODE_2
4 CATHODE_3
5 CATHODE_4
6 CATHODE_5
7 CATHODE_6
8 CATHODE_7
9 CATHODE_8
10 CATHODE_9
CATHODE_0
CATHODE_1
CATHODE_2
CATHODE_3
CATHODE_4
CATHODE_5
CATHODE_6
CATHODE_7
CATHODE_8
CATHODE_9
Anode Header (Right)
GND
J1
ANODE_DOTS
ANODE_TUBE_1
ANODE_TUBE_2
ANODE_TUBE_3
ANODE_TUBE_4
ANODE_TUBE_5
ANODE_TUBE_6
WS2812
ANODE_LEFT_DOTS
ANODE_TUBE_1
ANODE_TUBE_2
ANODE_TUBE_3
ANODE_TUBE_4
ANODE_TUBE_5
ANODE_TUBE_6
WS2812B
+5V
Support Headers
J3
J4
Additional Header
J2
CTDCR
CTDCL
DOT_R
RIGHT_DECIMAL_POINT
LEFT_DECIMAL_POINT
ANODE_RIGHT_DOTS
@Dimasssius
@DIY_Alliance

Anode Tubes Drivers



Decoder - K155ID1



Anode Tubes Resistor

Данным резистором можно регулировать яркость свечения индикаторов и разделительных точек, чем меньше сопротивление, тем выше яркость (и выше ток, что влияет на срок жизни индикаторов)

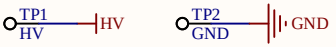
Варианты:
1. 33 кОм
2. 10 кОм (рекомендуемый)
3. 4.7 кОм
4. 3.3 кОм



Optoisolators Cathode Resistor

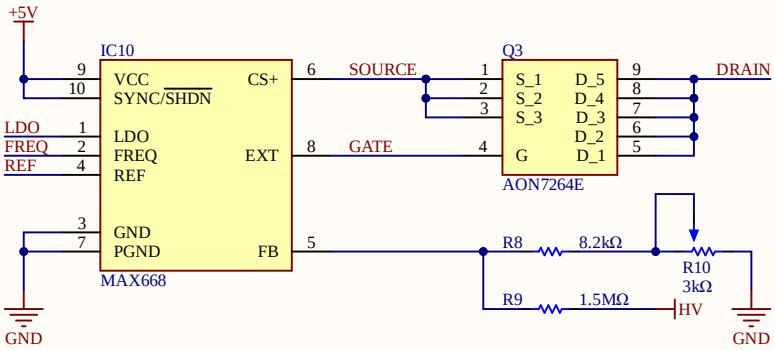


Test Point - High Voltage

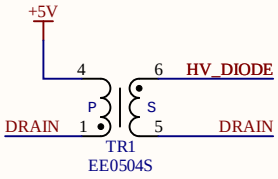


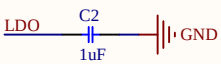
@Dimasssius
@DIY_Alliance

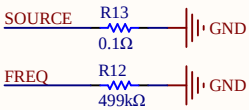
High Voltage Generator

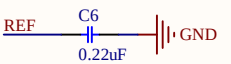


Transformer 1:10

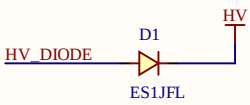


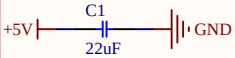


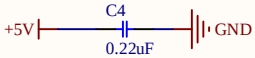


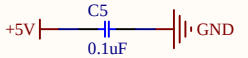


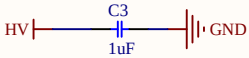
+180V

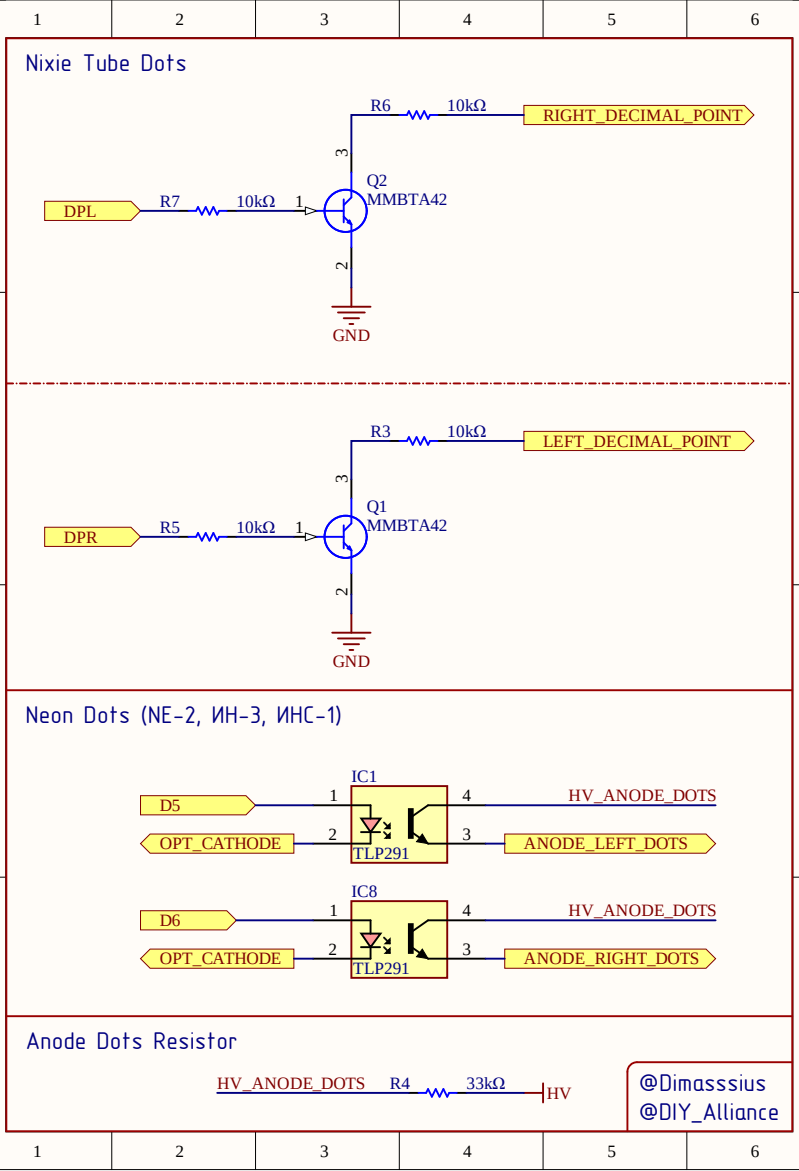


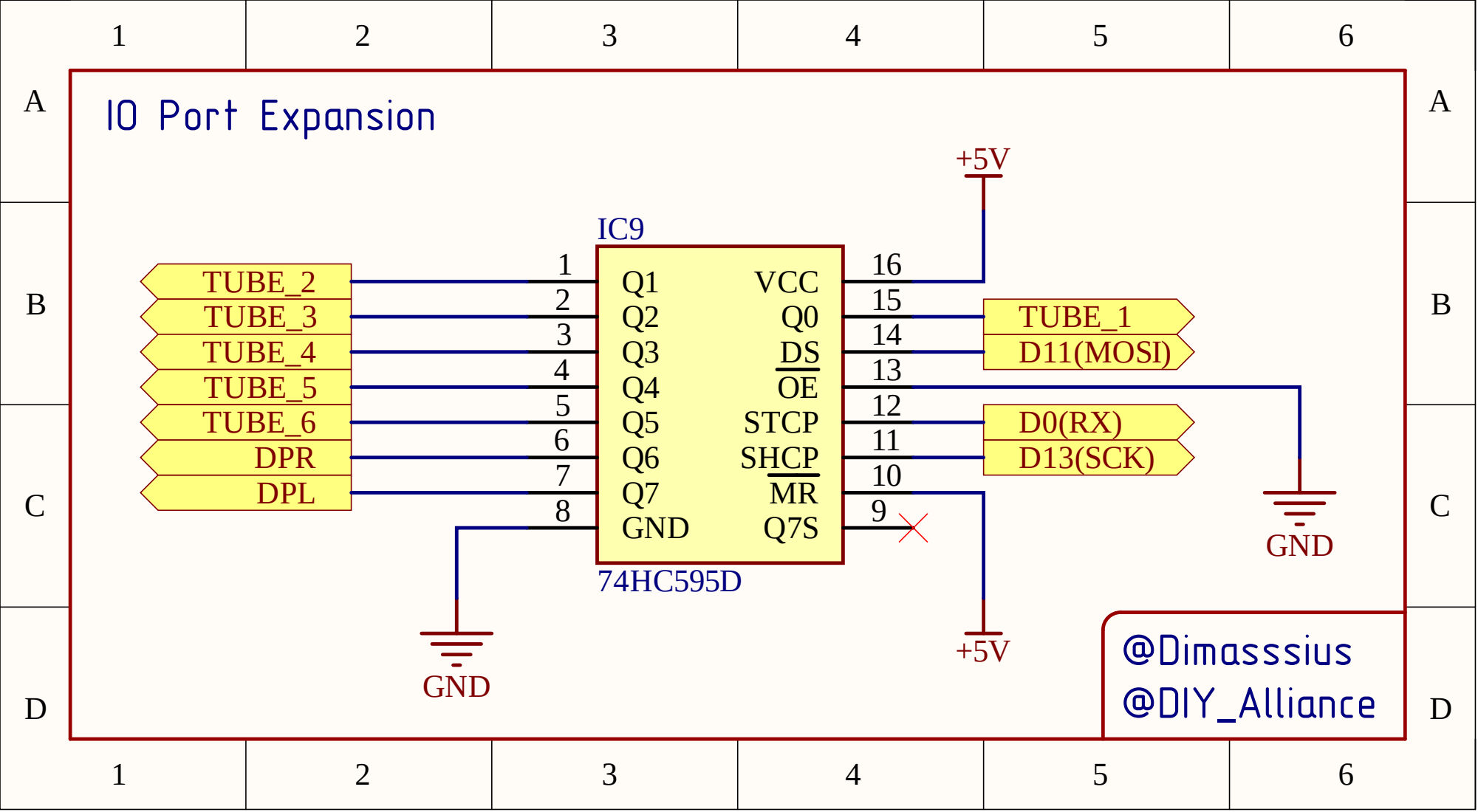


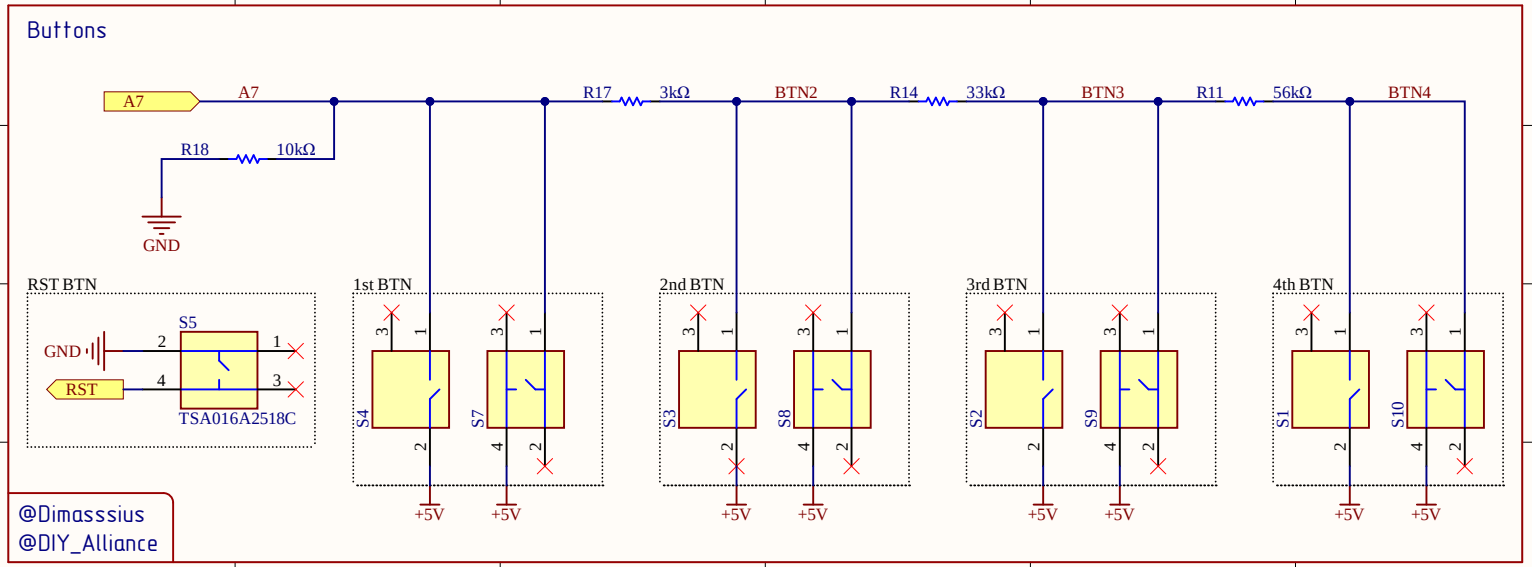




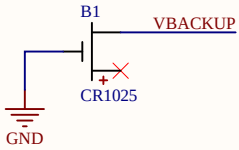



Nixie Tube Dots
R6
10kΩ
RIGHT_DECIMAL_POINT
DPL
R7
10kΩ
Q2
MMBTA42
GND
R3
10kΩ
LEFT_DECIMAL_POINT
DPR
R5
10kΩ
Q1
MMBTA42
GND
Neon Dots (NE-2, ИН-3, ИНС-1)
IC1
D5
OPT_CATHODE
TLP291
HV_ANODE_DOTS
ANODE_LEFT_DOTS
IC8
D6
OPT_CATHODE
TLP291
HV_ANODE_DOTS
ANODE_RIGHT_DOTS
Anode Dots Resistor
HV_ANODE_DOTS
R4
33kΩ
HV
@Dimasssius
@DIY_Alliance

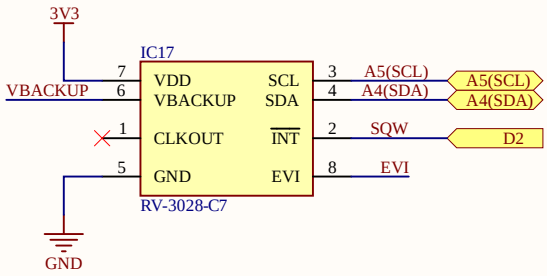

IO Port Expansion
IC9
74HC595D
TUBE_2
TUBE_3
TUBE_4
TUBE_5
TUBE_6
DPR
DPL
1 Q1
2 Q2
3 Q3
4 Q4
5 Q5
6 Q6
7 Q7
8 GND
16 VCC
15 Q0
14 DS
13 OE
12 STCP
11 SHCP
10 MR
9 Q7S
TUBE_1
D11(MOSI)
D0(RX)
D13(SCK)
+5V
+5V
GND
GND
@Dimasssius
@DIY_Alliance


Buttons
A7
A7
R18
10kΩ
GND
R17
3kΩ
BTN2
R14
33kΩ
BTN3
R11
56kΩ
BTN4
RST BTN
S5
GND
RST
TSA016A2518C
1st BTN
S4
S7
2nd BTN
S3
S8
3rd BTN
S2
S9
4th BTN
S1
S10
+5V
@Dimasssius
@DIY_Alliance

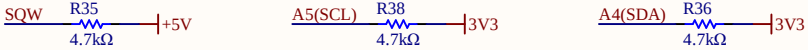
Battery Holder



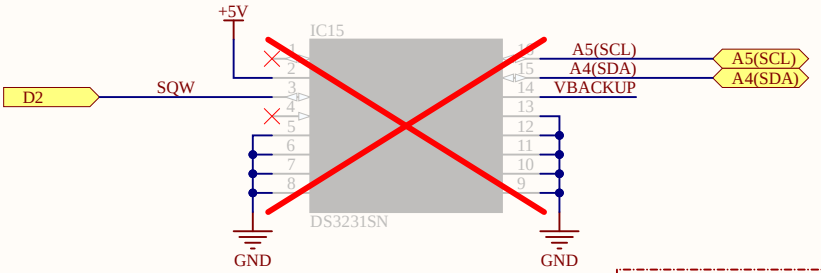
RV-3028-C7 - Variant #1



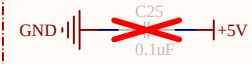
Pull-Ups

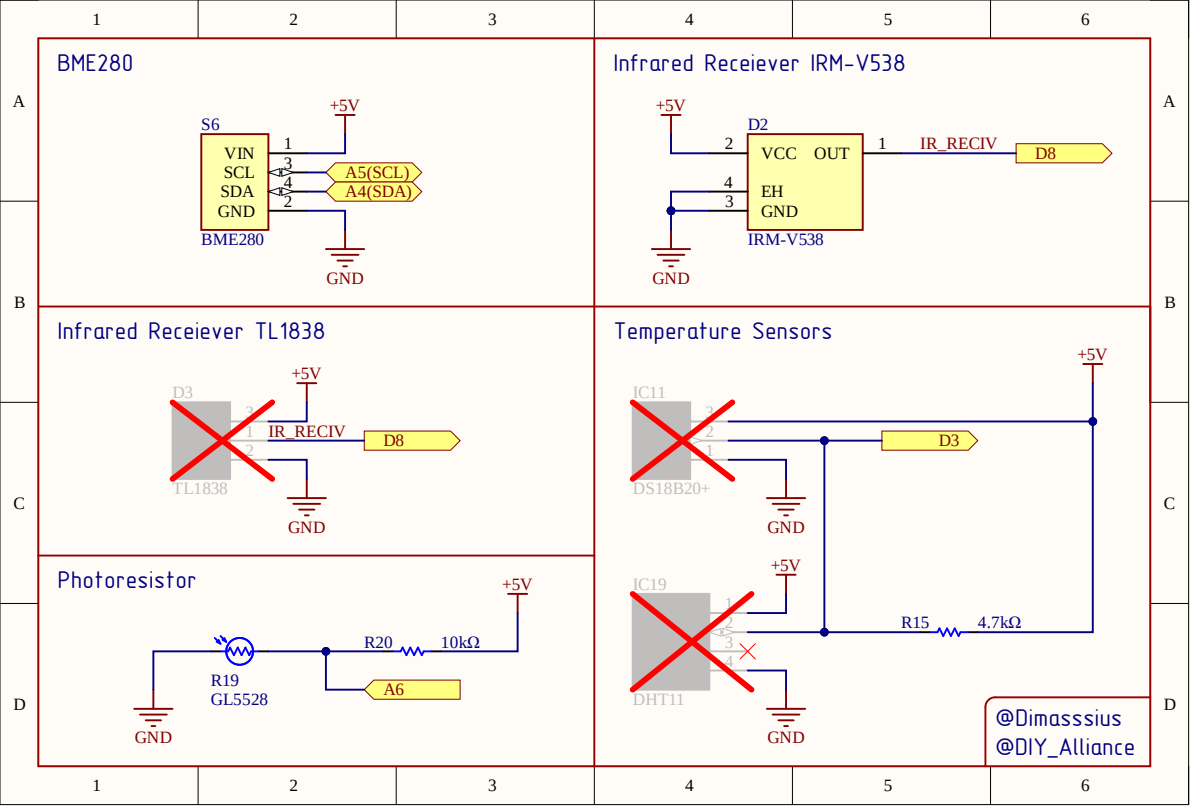


DS3231 - Variant #2



@Dimasssius
@DIY_Alliance

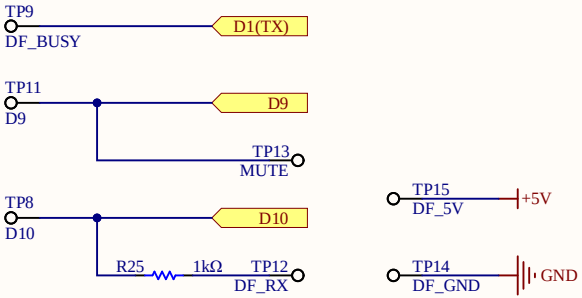



1
2
3
4
5
6
A
B
C
D
BME280
S6
VIN
SCL
SDA
GND
BME280
+5V
A5(SCL)
A4(SDA)
GND
Infrared Receiever IRM-V538
+5V
D2
VCC
OUT
EH
GND
IRM-V538
IR_RECIV
D8
GND
Infrared Receiever TL1838
D3
TL1838
+5V
IR_RECIV
D8
GND
Temperature Sensors
IC11
DS18B20+
+5V
D3
GND
IC19
DHT11
+5V
R15
4.7kΩ
GND
Photoresistor
+5V
R20
10kΩ
R19
GL5528
A6
GND
@Dimasssius
@DIY_Alliance

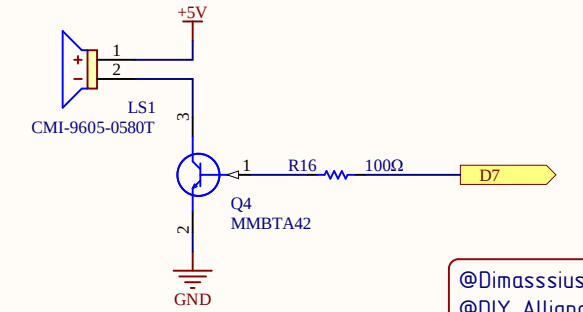
RDA5807M - Radio Tuner Connection Points



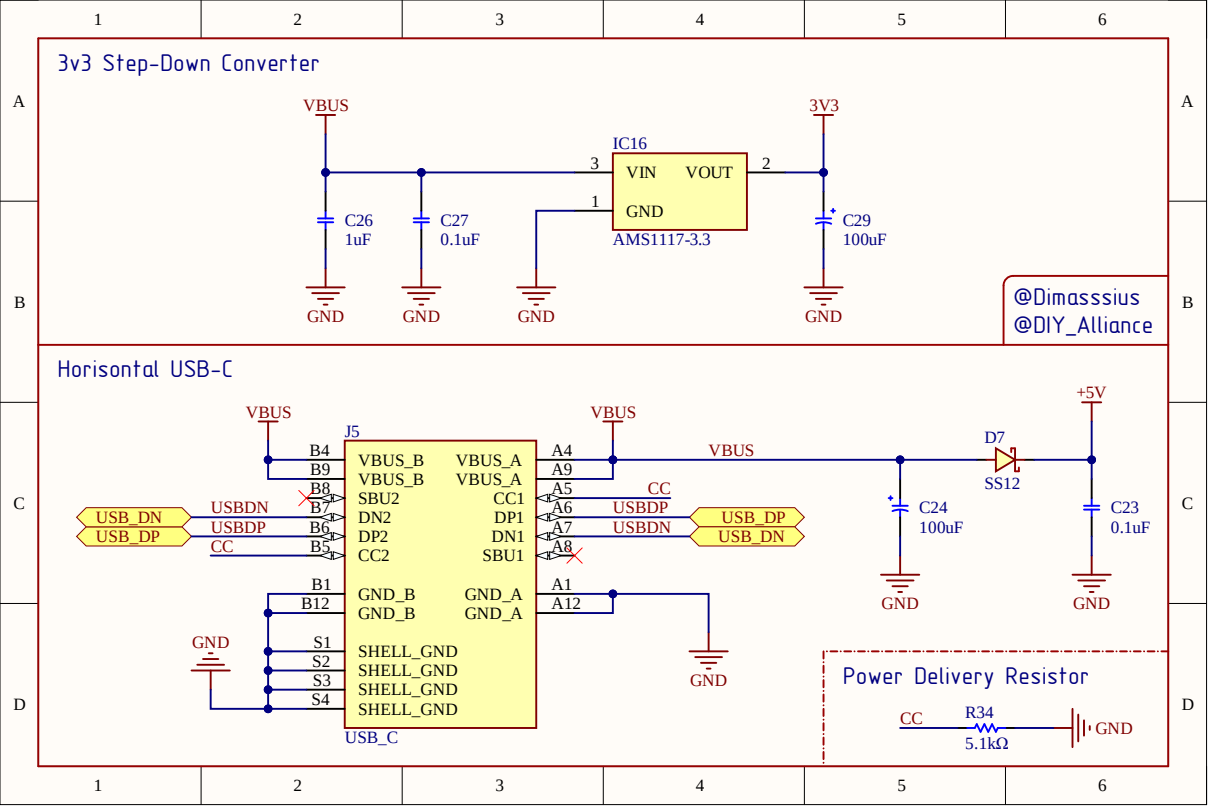
DFPlayer Connection Points

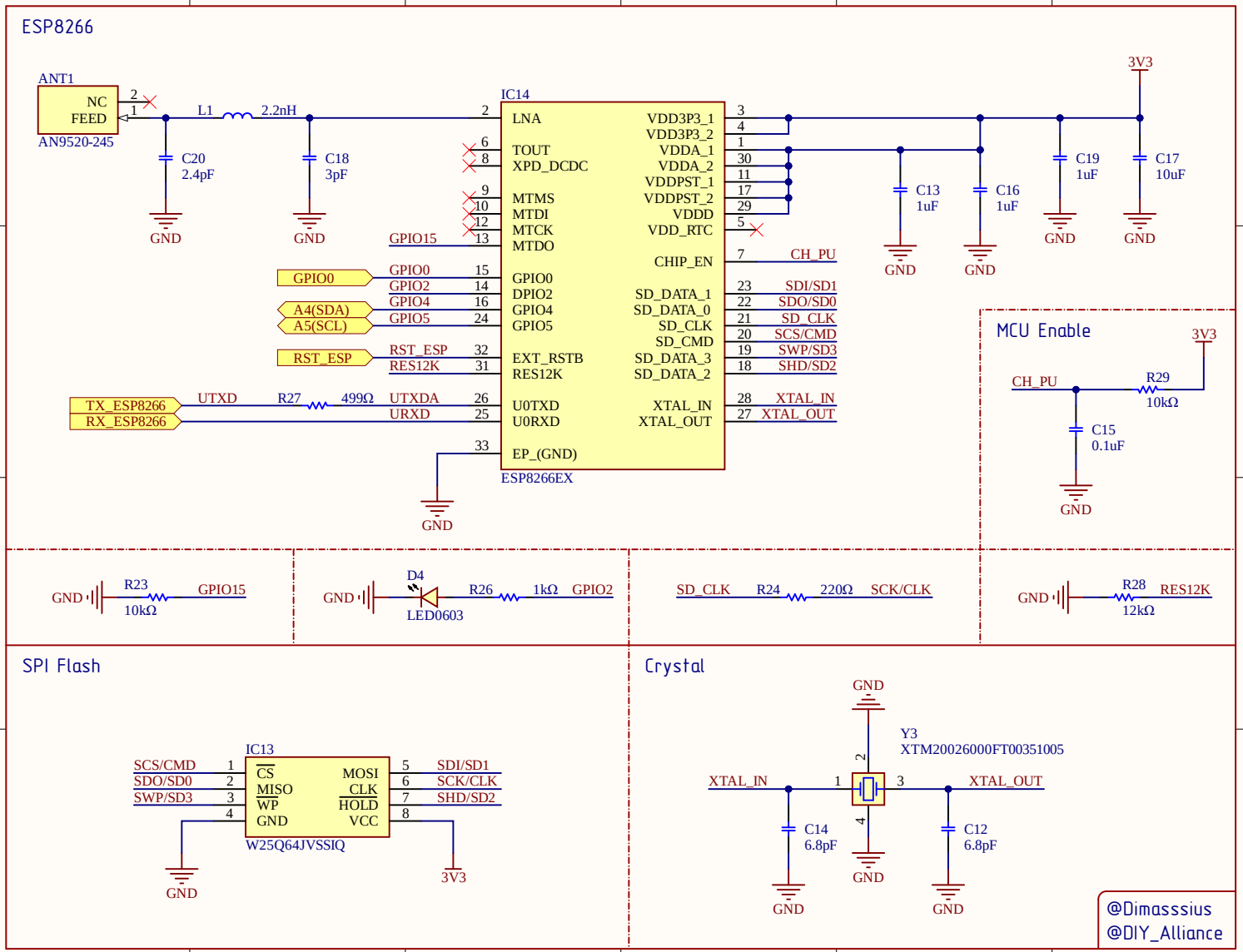


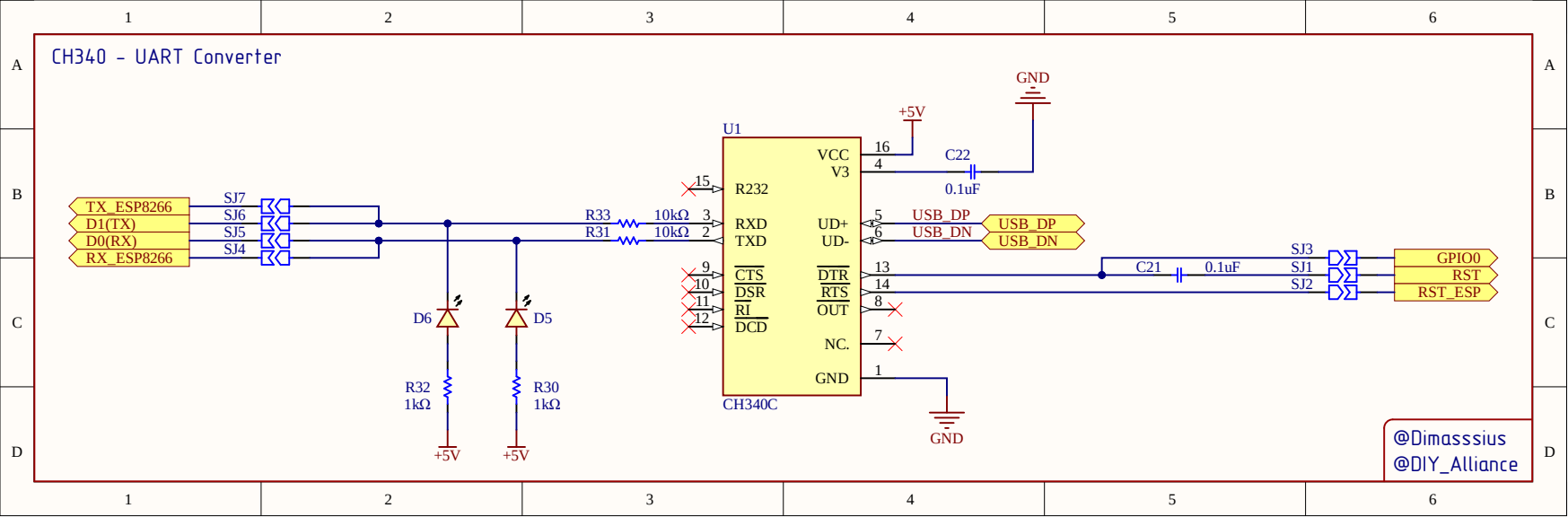
Buzzer



@Dimasssius
@DIY_Alliance


3v3 Step-Down Converter
VBUS
3V3
IC16
VIN
VOUT
GND
AMS1117-3.3
C26
1uF
C27
0.1uF
C29
100uF
@Dimasssius
@DIY_Alliance
Horisontal USB-C
J5
VBUS_B
VBUS_A
SBU2
CC1
DN2
DP1
DP2
DN1
CC2
SBU1
GND_B
GND_A
SHELL_GND
USB_C
USB_DN
USB_DP
USBDN
USBDP
CC
D7
SS12
+5V
C24
100uF
C23
0.1uF
Power Delivery Resistor
R34
5.1kΩ


ESP8266
ANT1
NC
FEED
AN9520-245
L1
2.2nH
C20
2.4pF
C18
3pF
GND
IC14
LNA
TOUT
XPD_DCDC
MTMS
MTDI
MTCK
MTDO
GPIO0
DPIO2
GPIO4
GPIO5
EXT_RSTB
RES12K
U0TXD
U0RXD
EP_(GND)
ESP8266EX
VDD3P3_1
VDD3P3_2
VDDA_1
VDDA_2
VDDPST_1
VDDPST_2
VDDD
VDD_RTC
CHIP_EN
SD_DATA_1
SD_DATA_0
SD_CLK
SD_CMD
SD_DATA_3
SD_DATA_2
XTAL_IN
XTAL_OUT
3V3
C13
1uF
C16
1uF
C19
1uF
C17
10uF
CH_PU
SDI/SD1
SDO/SD0
SD_CLK
SCS/CMD
SWP/SD3
SHD/SD2
XTAL_IN
XTAL_OUT
GPIO15
GPIO0
GPIO2
GPIO4
GPIO5
A4(SDA)
A5(SCL)
RST_ESP
RES12K
TX_ESP8266
RX_ESP8266
UTXD
R27
499Ω
UTXDA
URXD
MCU Enable
R29
10kΩ
C15
0.1uF
R23
10kΩ
D4
LED0603
R26
1kΩ
R24
220Ω
SCK/CLK
R28
12kΩ
SPI Flash
IC13
SCS/CMD
SDO/SD0
SWP/SD3
CS
MISO
WP
GND
MOSI
CLK
HOLD
VCC
SDI/SD1
SCK/CLK
SHD/SD2
W25Q64JVSSIQ
Crystal
Y3
XTM20026000FT00351005
C14
6.8pF
C12
6.8pF
@Dimasssius
@DIY_Alliance


CH340 - UART Converter
TX_ESP8266
D1(TX)
D0(RX)
RX_ESP8266
SJ7
SJ6
SJ5
SJ4
R33
10kΩ
R31
10kΩ
D6
D5
R32
1kΩ
R30
1kΩ
+5V
+5V
U1
R232
RXD
TXD
CTS
DSR
RI
DCD
VCC
V3
UD+
UD-
DTR
RTS
OUT
NC.
GND
CH340C
+5V
C22
0.1uF
GND
USB_DP
USB_DN
USB_DP
USB_DN
C21
0.1uF
SJ3
SJ1
SJ2
GPIO0
RST
RST_ESP
GND
@Dimasssius
@DIY_Alliance